

## 1. Feature Application: Schedule Conflict Detection System

### Feature Description

The Schedule Conflict Detection System automatically checks the academic schedule to detect conflicts such as two classes assigned to the same room at the same time, overlapping instructor schedules, or incorrect classroom assignments.

When a conflict is detected, the system flags the issue so administrators can correct the schedule before it is finalized. This ensures accurate classroom scheduling and prevents double booking.

### How to Use Feature

Administrator uploads or inputs the class schedule into the system.

The system scans the schedule database for conflicts.

If a conflict is detected, the system flags the schedule entry.

The administrator reviews the conflict details (room, class, time).

The administrator updates the schedule to resolve the issue.

## 2. Feature Description

The Smart Room and Building Status Indicator System integrates physical color indicators outside rooms with digital building colors on the campus map app using a unified color scheme. Physical indicators show room status, while the app highlights buildings based on users' personal schedules, helping students visually identify building locations and classroom availability within the campus.

Physical Layer: Color-coded lights outside rooms (classrooms, labs, faculty rooms, function halls) display current room status.

Digital Layer: Buildings on the campus map are color-coded based on the user's class schedule through a custom UI integrated with Google Maps as a visual map viewer only.

### Unified Color Coding System

Physical Room Indicators & Digital Building Indicators:

-  Red – Occupied/Current active class location
-  Green – Free/Available or Upcoming class today
-  Yellow – Reserved/Upcoming class within 15-30 minutes
-  White – Default/No scheduled classes

### How to Use the Feature

#### Physical Room Indicators:

- System connects to the school's class schedule database
- Indicator turns  Yellow 15-30 minutes before class (Reserved)
- Turns  Red when class begins (Occupied)
- Turns  Green when class ends (Free)
- Administrators can manually override status when needed

#### Digital Building Indicators:

- User taps buildings to view class details, room number, and instructor
- The application displays building locations only as a visual campus reference
- Google Maps integration is limited to displaying campus buildings through a custom UI
- The system does NOT provide real-time GPS tracking, route calculation, live navigation, or turn-by-turn directions
- Selecting a building opens the Floor Plan UI
- The Floor Plan UI displays the building interior separated by floors
- Students can view room IDs, hallways, classrooms, laboratories, offices, stairways, and facilities within the selected floor
- Building colors update automatically throughout the day based on schedule information

#### System Integration:

- Selecting a building shows room-level indicators within the Floor Plan UI
- Physical and digital statuses are synchronized in real-time
- The map serves only as a visual reference from the student's previous classroom or known location
- The primary navigation function of the application is the interactive Floor Plan UI system

## 2. Feature: Today's Campus Events Ticker

The Campus Events Ticker displays a real-time scrolling banner and a full list of today's campus events including the time, title, and category. Students, faculty, and visitors can quickly see what's happening on campus without needing to check bulletin boards or group chats. This feature directly addresses: How can students and campus members stay informed about daily events and activities in a quick and accessible way?

#### How to Use the Feature:

Open the Smart Campus app.

The events ticker automatically scrolls at the top of the screen.

Scroll to the Events tab for the full list.

Tap a category filter (Social, Food, Academic, Wellness, Arts, Sports) to narrow results.

View the event time, title, and location at a glance.

3. Feature Description: The Class Schedule Notification System automatically informs students, faculty, and administrators about class schedule updates, including class cancellations, rescheduling, room changes, and make-up classes. This ensures users always have accurate and up-to-date schedule information, preventing confusion and unnecessary campus travel.

This feature directly addresses:

How will students know when there are no classes due to interruptions, holidays, or school events?

How to Use the Feature:

User logs in to the mobile application.

The system syncs with the official academic schedule database.

When a class schedule is:

Cancelled

Rescheduled

Moved to another room

Affected by holidays or events

a notification is automatically generated.

Users receive real-time alerts based on their enrolled or assigned classes.

Users can tap the notification to view updated class details.

4. Feature Description:

The "Building & Floor Selector" allows users to quickly choose a specific building and floor within the campus to view available rooms and building layouts efficiently. This feature simplifies indoor visual navigation by reducing map clutter and helping users focus only on the relevant areas, making it easier to locate classrooms, offices, and facilities without confusion.

This feature directly addresses:

How will users quickly find rooms in large campuses with multiple buildings and floors?

How to Use the Feature:

1. User opens the Smart Campus mobile application.
2. The application displays a campus map using Google Maps as a visual map viewer only with a custom UI overlay.
3. The map does NOT use real-time GPS navigation, route tracking, or live location monitoring.
4. User selects a recognized building from the map or building list (e.g., Engineering Building, Main Building).
5. Once a building is selected, the Floor Plan UI automatically opens.
6. User selects a floor (e.g., Ground Floor, 2nd Floor, 3rd Floor).
7. The system displays an interactive visual floor plan showing only the selected floor.

8. Available rooms and facilities on that floor are highlighted.
9. User taps a room to view room details, room IDs, and facility information.
10. The Floor Plan UI serves as the primary navigation and room reference system of the application.

#### 5. Feature Description: Feedback and Rating System

The Feedback and Rating System allows students, faculty, staff, and visitors to rate campus buildings and provide comments about facilities such as cleanliness, accessibility, lighting, comfort, and overall experience.

The system collects ratings (1–5 stars) and written feedback, which can be reviewed by administrators to improve campus services and infrastructure.

This feature directly addresses:

How can users conveniently and privately provide feedback about campus facilities to help improve the campus facilities and infrastructure?

How to Use the Feature:

Open the Smart Campus app.

Scroll to the Feedback & Rating section.

Select a building from the building list.

Choose a rating (1 to 5 stars).

Write a comment or suggestion.

Tap Submit.

The feedback is saved and can be viewed by the school administrators.

#### 6. Feature Application: Floor Plan Builder

##### Feature Description

The Floor Plan Builder is an interactive editor that allows administrators and facility managers to design and visualize the physical layout of buildings and rooms. Using drawing tools such as lines, shapes, and connectors, users can create accurate floor plans that represent classrooms, offices, hallways, and other spaces within a building. The editor provides a canvas-based workspace where elements can be placed, resized, and labeled to reflect real-world room arrangements. This ensures that room assignments are grounded in actual physical structures, supporting better space management and scheduling decisions.

##### How to Use Feature

1. Administrator opens the Floor Plan Builder from the facility management module.
2. The user selects a building to design or creates a new building layout from scratch.
3. Using the editor toolbar, the user draws rooms and spaces using shapes (rectangles, polygons) and defines walls using line tools.
4. Each shape or room is labeled with a room name, number, or type (e.g., Classroom, Lab, Office).
5. The user connects rooms and spaces using connectors or path lines to represent hallways and entry points.

6. The completed floor plan is saved and linked to the scheduling system so room assignments reflect the actual building layout.
7. Administrators can update or edit the floor plan at any time to reflect physical changes in the building.

## 8.Feature Application: Class Schedule Integration

### Feature Description

The Class Schedule Integration feature connects the floor plan and room management system directly with the academic scheduling database. This integration ensures that every room displayed in the Floor Plan Builder is automatically synced with class schedules, instructor assignments, and time slots. When a class is scheduled in a specific room, the system reflects that assignment within the building and floor plan interface, giving administrators a unified view of room availability and occupancy across the entire building.

A key functionality of this feature is the ability for students to upload their Bicol University Certificate of Registration (COR), which contains their official class schedules. Once uploaded, the document is sent to an AI engine that reads, extracts, and converts the schedule data from the COR into a structured digital class schedule. This converted schedule is then automatically synced to the student's account and integrated with the other features of the application, such as the Floor Plan Builder, Schedule Conflict Detection System, and room assignment tools.

The application may integrate Google Maps only as a visual campus map viewer with a custom user interface. The system will not use real-time GPS navigation, live tracking, or route calculation functions. Instead, students can select a campus building from the map to open the Floor Plan UI, which acts as the application's primary navigation and room reference feature.

The Floor Plan UI allows students to view the building interior separated by floors and identify classrooms, laboratories, offices, hallways, stairways, and room IDs visually based on their class schedules and previous classroom locations.